

Siddharamayya Mathapati

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PROFESSIONAL SUMMARY

Senior AI Engineer with 5+ years of experience building production-grade LLM applications, multi-agent systems, and RAG pipelines across banking and enterprise domains. Skilled in LangChain, LangGraph, AWS Bedrock, and MLOps with expertise in fine-tuning LLMs using QLoRA and LoRA, prompt engineering, model evaluation, and deploying AI workloads on AWS, GCP, and Azure using Docker and CI/CD pipelines.

EDUCATION

Acharya Institute of Technology
Master's Computer Applications | CGPA: 7.2
B K College, Chikodi
Bachelor Computer Application | CGPA: 6.8

Bangalore, India
2018 - 2020
Chikodi, India
2015 - 2018

SKILLS SUMMARY

- **Languages:** Python, Bash/Shell Scripting
- **Frameworks:** LangChain, LangGraph, LangSmith, FastAPI, Django, PySpark, TensorFlow, HuggingFace
- **AI/ML:** LLMs, RAG, Prompt Engineering, Model Evaluation, NLP, RLHF, QLoRA, LoRA, MLOps, Multi-Agent Systems, Model Serving, Gen AI
- **Tools:** Git, Docker, CI/CD, VS Code, MLflow, Kubernetes, AWS SageMaker, FAISS, ChromaDB, Ollama
- **Cloud & APIs:** AWS (Bedrock, SageMaker), GCP, Azure, OpenAI API
- **Databases:** MySQL, PostgreSQL, MongoDB Atlas, VectorDB, ChromaDB, FAISS

WORK EXPERIENCE

Senior Artificial Intelligence Engineer | Maveric Systems Limited **June 2025 - Present**
Financial technology company specializing in banking transformation and AI-powered solutions.

- Built a production-grade multi-agent AML investigation platform using **LangGraph** and **AutoGen** with an 8-node pipeline ingesting real-time Kafka alert streams and auto-generating SAR reports.
- Integrated **AWS Bedrock** (Mistral, Titan Embeddings) with MongoDB Atlas Vector Store and **LangSmith** tracing; implemented PII redaction and schema-validated JSON outputs for compliance.
- Designed **PRISM**, an AI pipeline converting regulatory PDFs into executable DMN/DRL rules via Apache KIE/Drools using a 7-stage async LLM workflow with OpenTelemetry observability.
- Architected **CodeLens**, a microservices AI code analysis platform with 7 FastAPI agent services using Tree-sitter AST parsing, Ollama-hosted LLMs, deployed via **Docker** and **CI/CD**.
- Built a spec-compliant **MCP** server exposing 10+ banking tools over HTTP, WebSocket, and STDIO, integrated with AutoGen multi-agent orchestration using Qwen2.5 and Gemma models.
- Implemented LLM observability using LangSmith with A/B **prompt experimentation**, statistical **evals**, human feedback annotation, and prompt versioning via LangSmith Hub.
- Developed an AI loan delinquency prediction system with 0-999 **risk scoring** delivering context-aware repayment strategies and multi-channel communication workflows.

Senior Software Engineer | Capgemini Technology Services Limited **May 2021 - April 2025**
Multinational technology and consulting firm delivering large-scale AI and digital transformation solutions.

- Fine-tuned LLMs using **QLoRA** and **LoRA** for domain-specific adaptation in finance, healthcare, and cybersecurity, reducing memory overhead while maintaining model accuracy.
- Designed and deployed **RAG** applications integrating **FAISS** and **ChromaDB** vector stores with hybrid search for accurate, context-aware responses at scale.
- Applied **prompt engineering** and **model evaluation (evals)** frameworks to iteratively improve LLM accuracy and task alignment across multiple production deployments.
- Built real-time model serving pipelines using **TensorRT** and **ONNX**; managed deployments with **Docker**, **CI/CD** pipelines, and **Kubernetes** for low-latency inference.
- Deployed AI workloads on AWS, GCP, and Azure using **OpenAI API** and **HuggingFace** models; optimized cloud costs and ensured high availability across environments.
- Leveraged **RLHF** to align fine-tuned LLMs with human feedback; built distributed training loops across GPUs for large-scale model development using **MLflow** tracking.